Assignment 1 - Image Warping

This repository is my implementation of Assignment 1 for Digital Image Processing. The work includes basic geometric transformation and point-guided image deformation with an interactive Gradio interface.

Requirements

Install the required packages with:

```
python -m pip install -r requirements.txt
```

Running

Run the global transformation demo:

```
python run_global_transform.py
```

Run the point-guided deformation demo:

```
python run_point_transform.py
```

Work Summary

1. Basic Image Geometric Transformation

In `run_global_transform.py`, I implemented:

- scaling, rotation, translation, and horizontal flipping
- affine matrix composition in homogeneous coordinates
- center-based scaling and rotation
- manual inverse mapping and bilinear interpolation without calling existing geometric transform APIs

2. Point Guided Image Deformation

In `run_point_transform.py`, I implemented:

- point pair selection through the Gradio interface
- MLS affine deformation based on control points
- backward warping from target pixels to source pixels
- bilinear interpolation for smooth deformation results

MLS Algorithm Overview

Moving Least Squares (MLS) is a point-guided deformation method that builds a local transformation for each output pixel. For a pixel in the target image, the algorithm first measures its distances to all target control points and assigns larger weights to nearby points. It then computes weighted centroids of the target points and source points, estimates a local affine transform between the two centered point sets, and maps the pixel back to the source image. Because the transformation is computed locally and weighted by distance, pixels near edited control points deform more strongly, while far regions remain smooth and natural.

Results

Basic Transformation

`global_1.png` shows image scaling and rotation.

`global_2.png` shows image translation.

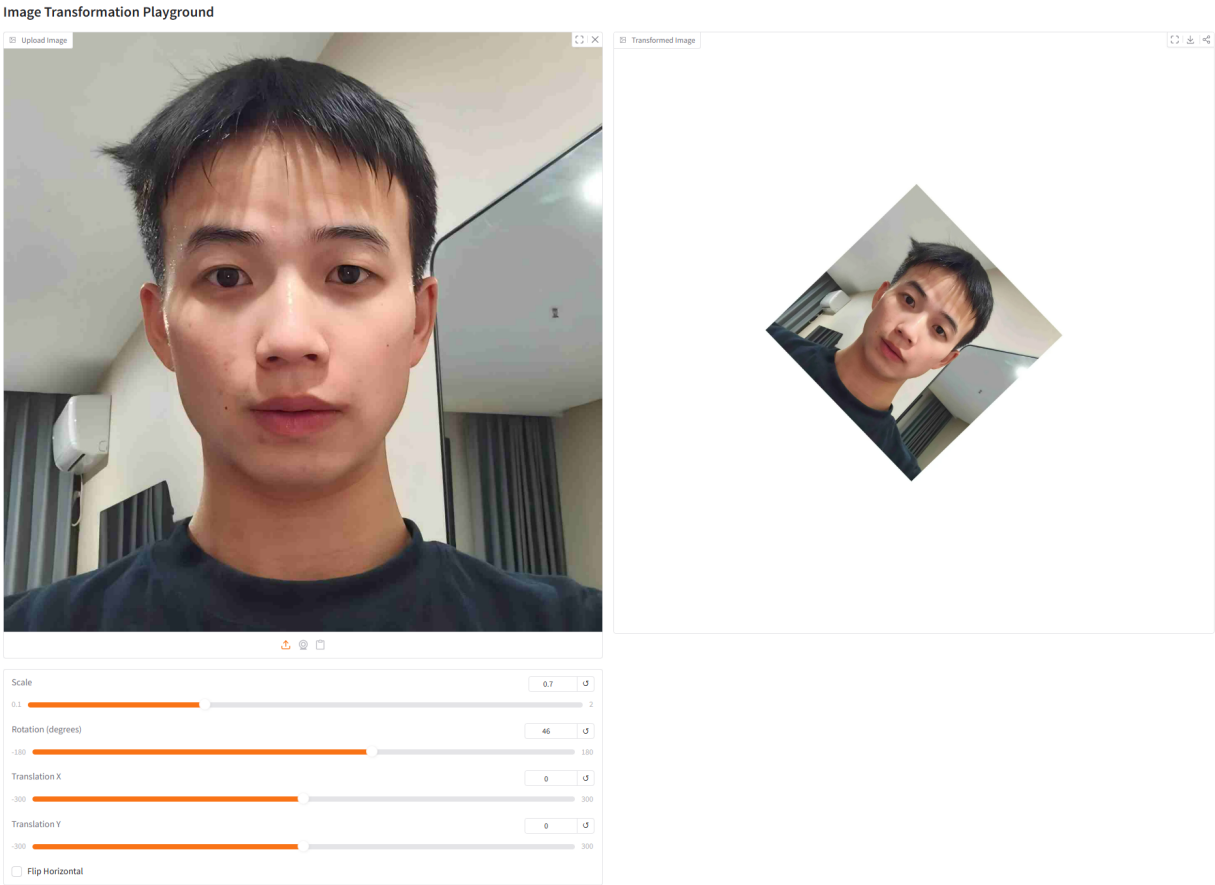

Figure 1: global transform scale and rotation

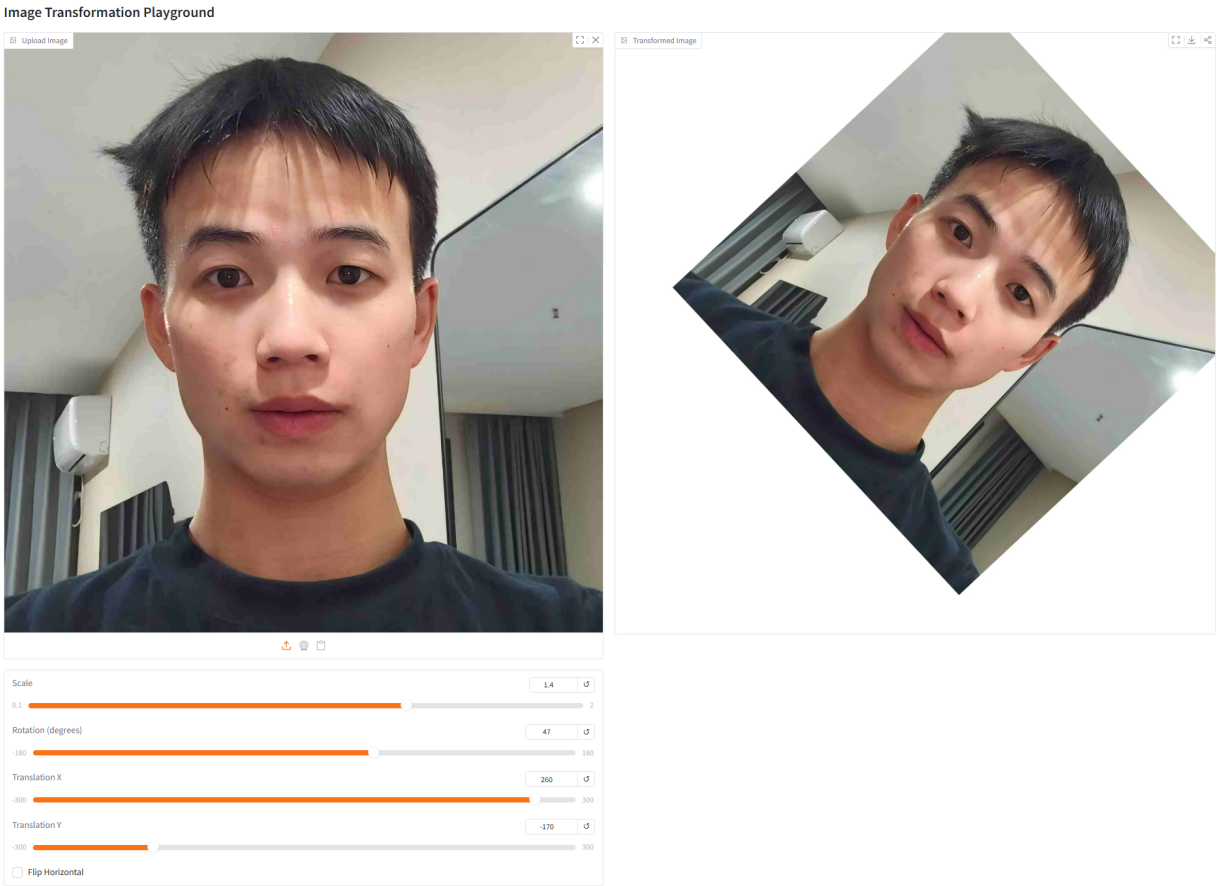

Figure 2: global transform translation

Point Guided Deformation

`point_1.png` shows the eyes moved outward and downward.

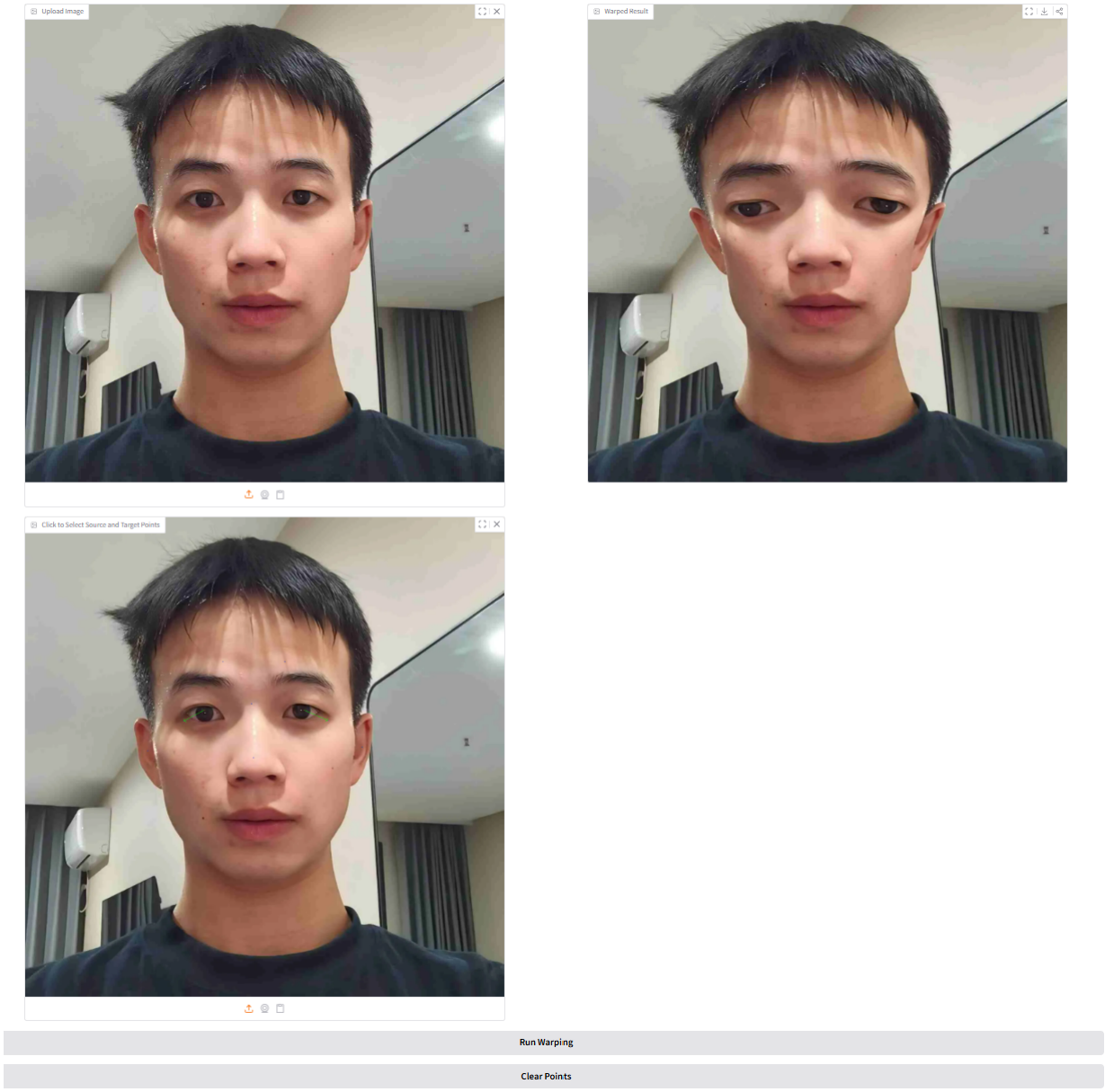

Figure 3: point guided deformation eyes

`point_2.png` shows the mouth corners lifted upward.

Conclusion

This assignment completes both global image transformation and local point-guided deformation. The global module supports combined affine operations, and the point-guided module produces natural local edits such as eye movement and mouth lifting.

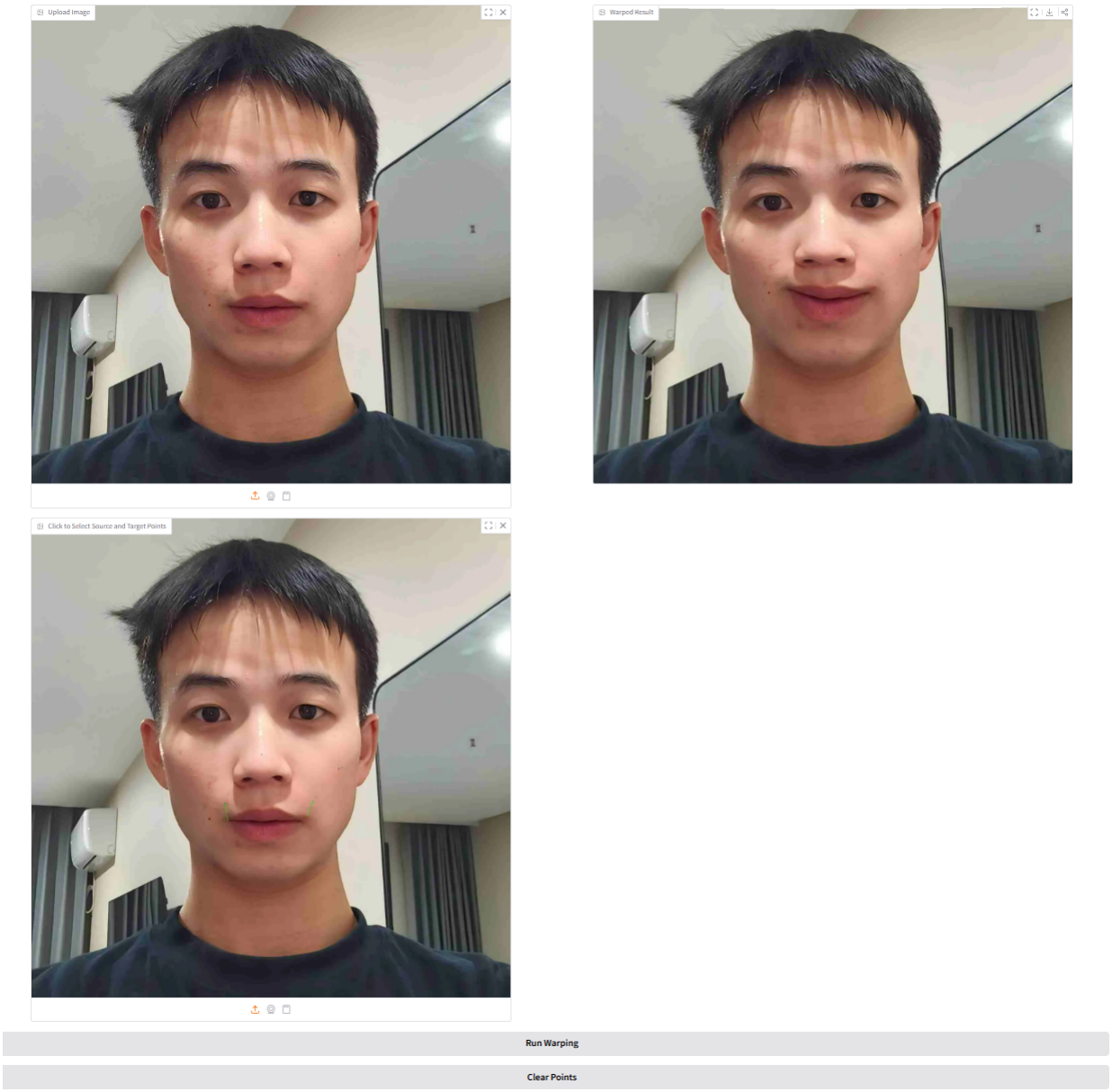

Figure 4: point guided deformation smile

References

- Teaching slides: 作业 01-ImageWarping.pptx
- Schaefer, McPhail, Warren. Image Deformation Using Moving Least Squares